

Dear examiner,

please find the criteria marks as well as teacher's comments below.

A: 4	B: 3	C: 3	D: 2	E: 5	Total: 17
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Criterion A: Presentation

A very comprehensive and easy to follow IA. The student successfully utilizes the mathematics required to investigate a real-life problem of his choice. Graphs, equations, tables and helpful diagrams are used effectively and referred to in his written explanations to aid understanding. The IA has an introduction with a clear aim, which is fully met. A conclusion summarizes the findings.

Criterion B: Mathematical communication

Appropriate and mostly consistent, relevant mathematical language is used. Unfamiliar key terms are defined and explained. A number of equations, diagrams, and tables are used. There are a few minor slips in terms of his mathematical communication (e.g. axes in graphs are not labeled, sloppy formulations like "rate of increase" on page 1, "subtract the integrals" instead of "subtract the area" on page 6", or confusing the terms "formula" and "expression" on page 13), however, they do not detract from the flow of the exploration.

Criterion C: Personal engagement

There is evidence of outstanding personal engagement. The student managed to really make this IA his own by applying relevant integration concepts to investigate an authentic real-life problem of his choice. The Math software GeoGebra is used effectively. This is a unique piece of work enthused by personal interest and mathematics was utilised creatively, driving the IA forward - the student's voice can very much be heard throughout.

Criterion D: Reflection

The student has done a very good job reflecting on his mathematical working at pretty much any stage which really helps to move the IA forward. Implications of the results are frequently discussed and analysed. The conclusion contains a summary of his findings, further reflection, and states two possible extensions. However, most reflection seen is not critical.

Criterion E: Use of mathematics

The student uses relevant mathematics commensurate with the level of the course. All mathematical arguments are clear and logical; all mathematics explored is error-free and an appropriate degree of accuracy is used. Thorough knowledge and understanding are demonstrated as the student supports all his mathematical working by clear and logical written explanations and diagrams applying the mathematics and "playing" with it. At times the student is a bit sloppy with his mathematical language but this shortcoming was penalized in criterion B.

An Investigation into the different cross sections of rain channels and their flood times.

Mathematics Analysis and Approaches HL

Internal Assessment

Student Code: hdv619

Table of Contents:

- 1. Introduction, Rationale and Aim**
- 2. Initial Scenario and Types of Rain Channels**
- 3. Rectangular Rain Channel**
- 4. Trapezoidal Rain Channel**
- 5. Quadratic Cross Section**
- 6. Semi-Circular Rain Channel**
- 7. Comparing Time Functions**
- 8. Generalising the Time Function**
- 9. Flood Times for Changing Rainfall Rates**
- 10. Conclusion**

1. Introduction:

As cities cover more and more of their natural ground with impermeable surfaces, the risk of flooding steadily increases, since rainwater has nowhere to go. Rain channels are therefore essential to prevent streets and sidewalks in cities from flooding and ensure that rainwater flows off streets and paths and away from people. While their shape and function appears simple, it is essential for any building project to use mathematics to calculate the dimensions of the rain channels, so that they do not take up too much space in the city, but still ensure streets don't flood in the case of heavy rainfall. Factors such as volume, initial depth of water, changing rates of rainfall and other factors must be considered in order to calculate how long it will take for a channel to overflow, or how much rainfall a channel can handle before flooding.

1.1 Rationale:

When I took a vacation in Tenerife, I noticed the abundance of rain channels on the sides of the roads. Tenerife faces a lot of rain throughout the year, and the rain channels ensure that the streets do not get damaged too much in periods of heavy rainfall. When it rained, I saw how the rain channels take up the rainwater that fell onto the street and the sidewalks. This led me to wonder: "How long would it take for the rain channel to fill up if it kept raining at this rate?" I also asked myself: "Could you formulate an equation, which takes all of these variables into consideration and calculates a time for how long it will take for the rain channel to fill up?"

1.2 Aim:

The aim of this investigation is to compare the flood times for different cross sections of a rain channel given set parameters, as well as constructing equations for each of the cross sections that can be used to find the time taken to flood the channels where different parameters can be changed, such as initial height of water, the width of a street draining into the channel, or the rate of increase of the rainfall rate as time passes.

2. Initial Scenario:

When in Tenerife, I wondered how long the rain channel would take to fill up under a certain rate of rainfall. To figure this out, different things must be considered, such as the size of the rain channel, its shape, the water already in it, and the rate of rainfall. For this investigation, we will assume that the water does not leave the rain channel and instead focus on the volumes of the channels and how their cross-sections relate to how quickly they fill under certain conditions. In

order to set a standard for all channels, we will set the max width and max depth of all channels to 20cm and 10cm respectively. This is a realistic size for a rain channel and will therefore allow for the calculations to be applicable to the real world. When looking through google, I found that typical rates of rainfall are around 10l per square meter per hour, so we will use this as our standard rainfall rate.

Research into different shapes of Rain Channels:

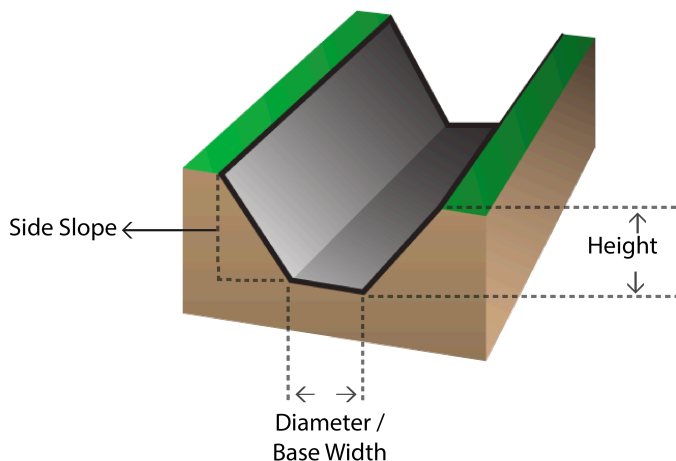
When looking through pictures on the internet, it seems that rain channels which have a semi-circular cross section seem to be the most common. As seen in figure 1, the channel is a semi-circle cut into the concrete. It is also the shape used in rain drains on roofs. It is probably widespread because it is easy to produce and its round channel means there are no points in the stone that wear away quicker than others.



Figure 1: A Semi-Circular Rain Channel (From <https://nylonhose.blogspot.com/>)

Another common shape for a rain channel is a rectangular cross section. It is also commonly used in rain drains on roofs. It is the simplest cross-sectional shape since it can be constructed with just two sheets of materials on a floor. A shape similar to the rectangular cross section is the trapezoidal rain channel, as seen in figure 2. It is probably used since it is like a rectangular rain

Trapezoidal Channel



channel but funnels water into the middle of the channel which might improve water flow. For the purpose of the investigation, I will also look at a rain channel which has a cross section represented by a quadratic function. This will allow me to compare the quadratic cross-section to the semi-circular cross section and will also provide an extra scenario where manual integration can be used.

Figure 2: An example of a trapezoidal rain channel (From innovyze.com)

3. Rectangular Rain Channel:

A typical shape of the rain channel, and the simplest design, is a rain channel with a rectangular cross section. To calculate the volume of this rain channel, we find the area of the cross section and multiply it by the length of the rain channel. The graph in figure 3 shows the dimensions of the cross section of the rain channel which is 20cm wide and 10 cm high. To find the cross-sectional area, we multiply the base by the height:

$$Area = 20 \times 10 = 200cm^2$$

To find the volume, we multiply the area by the length of 1 m, but since we are working with centimetres, we must convert 1 m to 100cm:

$$Volume = 200 \times 100 = 20,000cm^3$$

We now know that a rectangular straight rain channel with width 20cm, height 10cm and length 100cm has a volume of 20,000 cubic centimetres. We now need to find out how much volume of rainwater enters the rain channel in a specific time. Since we assume that rain is falling at a constant rate, we need to find the area of the top of the rain channel to deduce what area the rain falls onto, hence influencing how much water enters the rain channel. Assuming the rain channel is perfectly level, the horizontal area of the top is equal to the width multiplied by the length:

$$Area = 100 \times 20 = 2000cm^2$$

In the beginning, we assumed rainfall to be consistent at 10 litres per hour per square meter. We know that there are 1,000 cubic centimetres in a litre. We also know that one square meter has the same area as 10,000 square centimetres. To convert the rainfall rate to cubic centimetres per square centimetre per hour, we do the following conversions:

$$10lm^{-2}h^{-1} \rightarrow \frac{10 \times 1000}{10000} = 1cm^3cm^{-2}h^{-1}$$

To find the rainfall entering the rain channel, assuming only rain that falls directly onto the rain channel enters it, we multiply the rainfall rate by the area of the top of the rain channel:

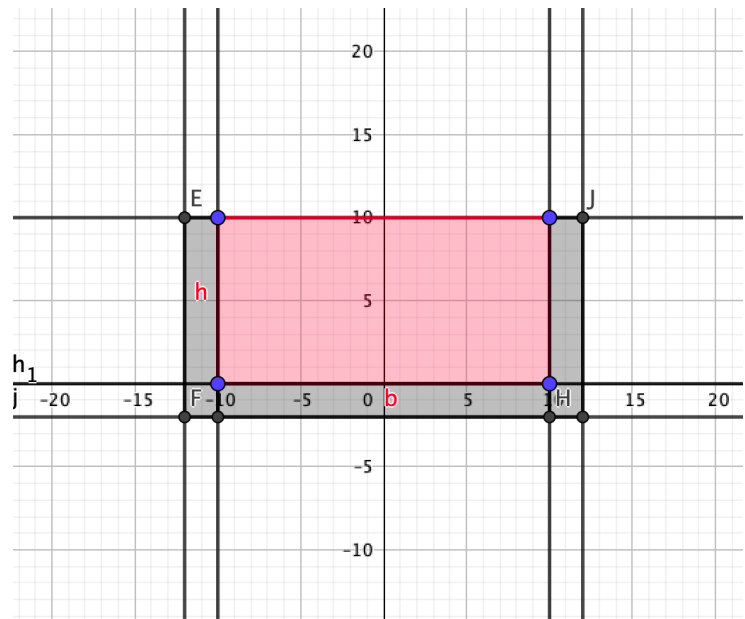


Figure 3: Cross-Section of a rectangular rain channel, the red area representing where water can flow into (Made by myself using GeoGebra)

$$1 \times 2000 = 2000 \text{ cm}^3 \text{ h}^{-1}$$

To now find the time it will take for the rain channel to fill up, we divide the volume by the fill rate:

$$\text{Time} = \frac{20000}{2000} = 10 \text{ h}$$

Therefore, for a straight rain channel, with rectangular cross section, which is 1 m long, 10 cm high and 20 cm wide, the time to fill it up from empty to full under a rainfall rate of 10l per square meter per hour is **10 hours**. This is surprisingly long considering we are using normal rainfall rates, however currently only the rain falling directly into the rain channel is filling it up.

Since the cross section is rectangular, changing the initial height of water in the rain channel does not change the width of the base of the rain channel. We will assume that rainwater is already in the rain channel, which is a likely scenario. Let us set the initial height of water to be 2.5 cm of water. This would mean the new cross section would be:

$$\text{Area} = 20 \times (10 - 2.5) = 20 \times (7.5) = 150 \text{ cm}^2$$

Our volume would be:

$$\text{Volume} = 150 \times 100 = 15,000 \text{ cm}^3$$

Finally, we have to divide the volume by the fill rate to find the time taken to fill up the remaining space in the rain channel. Note that the earlier fill rate we found can still be used here, since changing the height of the water in the rain channel has no effect on how much water enters the rain channel, which depends on the area of the top of the rain channel.

$$\text{Time} = \frac{15000}{2000} = 7.5 \text{ h}$$

As we can see from the 7.5 hour result, the time it takes to fill the rain channel decreases by the same percentage as the decrease height of empty volume in the rain channel due to increasing initial height of water. Therefore, we can generalise the function we use to find the time it takes to fill the rain channel in terms of the initial height of water. We know that the volume depends on the height, width and length of the channel, which are 10-h, 20 and 100 cm respectively. Furthermore, the rainfall rate is 2000 cubic centimetres per hour no matter what the initial height of water is. Hence, we arrive at the following equation:

$$Time = \left(\frac{(10 - h) \times 20 \times 100}{2000} \right) hours$$

$$Time = (10 - h) hours \rightarrow 0 < h < 10$$

which proves a linear relationship between time and h , where h is the initial height of water in the rain channel in centimetres.

4. Trapezoidal Rain Channel:

Another typical design for a rain channel is one which has a cross sectional area of the shape of a trapezoid. The purpose of this shape is to guide the flow of the water towards the centre of the rain channel. We will still use the same height for the rain channel, but the base width will be changed to half of the top width, in order to form the shape of a trapezoid, as seen in figure 4. To graph the cross section of this rain channel, we can use two piecewise functions which are reflected over the y -axis. The red area in figure 4 is enclosed by the two lines $y = 10$, $y = 0$ and the functions $f(x) = 2x - 10$ and $g(x) = -2x - 10$.

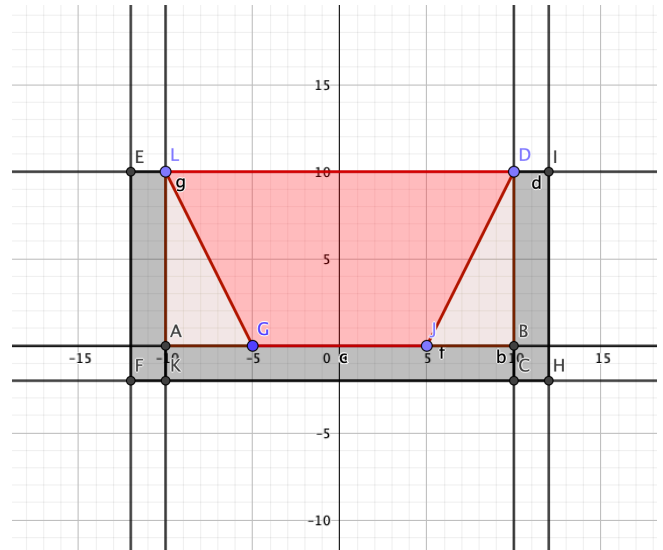


Figure 4: Cross-Section of a Trapezoidal Rain Channel (Made by myself using GeoGebra)

In order to find the area of the red region, we can use the formula for the area of a trapezoid:

$$A = \frac{a + b}{2} h$$

Since the piecewise functions intersect the $y = 0$ line at $(-5, 0)$ and $(5, 0)$, the first base length is 10 cm, while the second base length, in this case the width of the top, is 20cm. Using the height of 10cm, we get the following area:

$$Area = \frac{10 + 20}{2} \times 10 = 150cm^2$$

To find the volume, we multiply by the length of the channel, which is 100cm:

$$Volume = 150 \times 100 = 15,000cm^3$$

Once again, for the trapezoidal rain channel, water enters the channel at the same rate as before, as the area of the top of the channel has not been altered. Hence, we can calculate the time to fill the rain channel as before:

$$Time = \frac{15000}{2000} = 7.5h$$

This is notably less than the rectangular cross section due to the decrease in width of the base.

Now, if we want to find the time it takes for the trapezoidal rain channel to fill up depending on different initial heights of water in it, the simple formula for the area of the trapezoid won't be as effective, since the length lower base of the trapezoid will have to be repeatedly calculated for different initial fill heights. Instead, we can use an integral of the piecewise functions to find what area is filled by the water already in the rain channel and hence what area is still empty. For example, let us set the initial height of the water in the channel to 2.5cm. First, we can find the area of the empty cross section if our rain channel was rectangular, as in the first example, by multiplying the width of 20cm by the new height of the empty channel, 7.5 cm:

$$Area = 20 \times 7.5 = 150cm^2$$

We now shift the graph from figure 4 down by h , the initial height of water, so that the x -axis represents the height of the water already in the channel, as seen in figure 5. Since we are working with a trapezoidal cross section, we must subtract the integrals of the piecewise functions on either side, from the x -intercept to the max width of the rain channel, which is 10cm to either side, hence the x values 10 and -10. However, we only need to find the integral of the right function since the integral of the other function will be equal. Hence, we find the area by subtracting twice the integral of the function $f(x) = 2x - 10$ from $x = 6.25$ to $x = 10$ from the overall area 150:

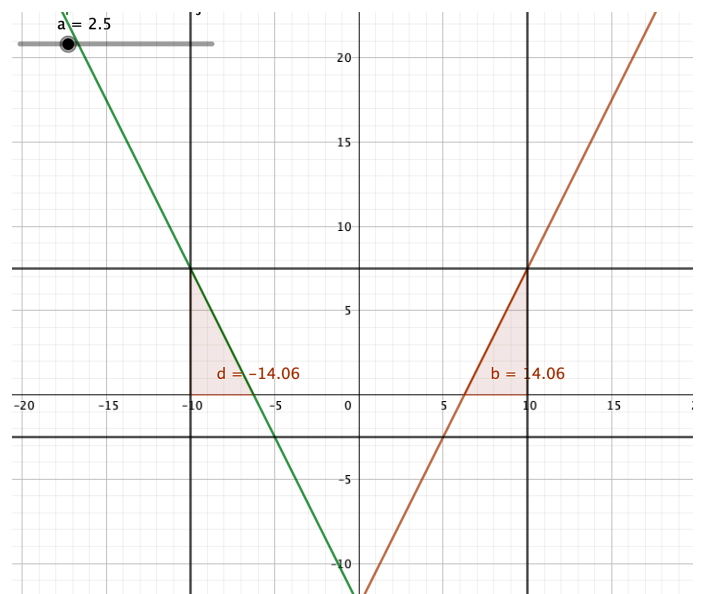


Figure 5: Changing trapezoidal area as initial height of water changes (Made by myself using GeoGebra)

$$Area = 150 - 2 \int_{6.25}^{10} (2x - 12.5) dx$$

Using technology to compute this integral, we arrive at the area:

$$Area = 121.875 \approx 122cm^2$$

Once again, we find the time taken to fill the rain channel by multiplying the cross sectional area by the 100 cm length and dividing by the 2000 cubic centimetres per hour rainfall rate:

$$Time = \frac{122 \times 100}{2000} = 6.1h$$

We can see that increasing the initial height of water does have a notable effect on the time taken to fill the channel as an initial height of 2.5cm already decreases the flooding time by over 1 hour.

However, this is only for a specific case. If we want to find a function for time in terms of initial height of water, we must integrate the function manually. The upper bound will remain 10 no matter what the initial height of water is. The lower bound, which is the zero or the y-intercept of the function, will change however as we change the initial height, h , of the water in the channel. We find the lower bound by solving the function for x when y is 0:

$$2x - (10 + h) = 0 \rightarrow x = \frac{10 + h}{2}$$

We can now substitute in the lower and upper bounds into the integral:

$$\int_{\frac{10+h}{2}}^{10} (2x - (10 + h)) dx$$

We can treat the $10+h$ in brackets as a constant, so we just multiply it by x . The $2x$ simply integrates to x squared since when differentiating x squared we get $2x$ back. We therefore arrive at the result:

$$\left[x^2 - (10 + h)x \right]_{\frac{10+h}{2}}^{10}$$

We can now substitute in the upper and lower bounds. To do this, we first replace x with the upper bound and then subtract the value we get when we replace x with the lower bound. This results in the following expression:

$$(-10h) - \left(\left(\frac{10+h}{2} \right)^2 - \left(\frac{(10+h)^2}{2} \right) \right)$$

To get from this expression to the area of the entire cross section, we subtract the integral from the area of the rectangle that has side lengths 10 and 10-h. Then we multiply that final area by 2 to get the full cross-sectional area of the rain channel:

$$\begin{aligned} Area &= 2 \times \left((10 \times (10-h)) - \left((-10h) - \left(\left(\frac{10+h}{2} \right)^2 - \left(\frac{(10+h)^2}{2} \right) \right) \right) \right) \\ Area &= (200 - 20h) - 2 \left((-10h) - \left(\left(\frac{100 + 20h + h^2}{4} \right) - \left(\frac{(100 + 20h + h^2)}{2} \right) \right) \right) \\ Area &= (200 - 20h) - \left((-20h) - \left(\left(50 + 10h + \frac{h^2}{2} \right) - (100 + 20h + h^2) \right) \right) \\ Area &= (200 - 20h) - \left(-20h - \left(-50 - 10h - \frac{h^2}{2} \right) \right) \\ Area &= (200 - 20h) - \left(50 - 10h + \frac{h^2}{2} \right) \end{aligned}$$

Finally, this formula simplifies to:

$$Area = \left(150 - 10h - \frac{h^2}{2} \right) cm^2$$

Now, we can easily find the time taken to fill up the rain channel depending on the height of water already in it, by simply multiplying by the 100cm length and dividing by the 2000 cubic centimetre per hour rainfall rate:

$$Time = \frac{100}{2000} \times \left(150 - 10h - \frac{h^2}{2} \right)$$

$$Time = \left(7.5 - \frac{1}{2}h - \frac{h^2}{40} \right) hours \rightarrow 0 < h < 10$$

5. Quadratic Cross Section:

The next cross-sectional shape for the rain channel we will look at is the cross section that is of the form of a quadratic equation. Even though this shape is unlikely for a real-life rain channel,

we can use it as a transition between the trapezoidal shape and the semi-circle. Once again, we will assume the top width to be 20 cm, and the height of the rain channel to be 10cm. The cross-section of the rain channel can be seen in the graph in figure 6.

Firstly, we will assume there is no water in the rain channel initially. To find the time taken to fill the rain channel, we first find the cross-sectional area of one side, since the cross section is symmetrical over the y-axis, and finding the area of one side is simpler. We do this by subtracting the integral of the graph from the rectangular area, and multiplying by 2 since we are only finding the area on one side.

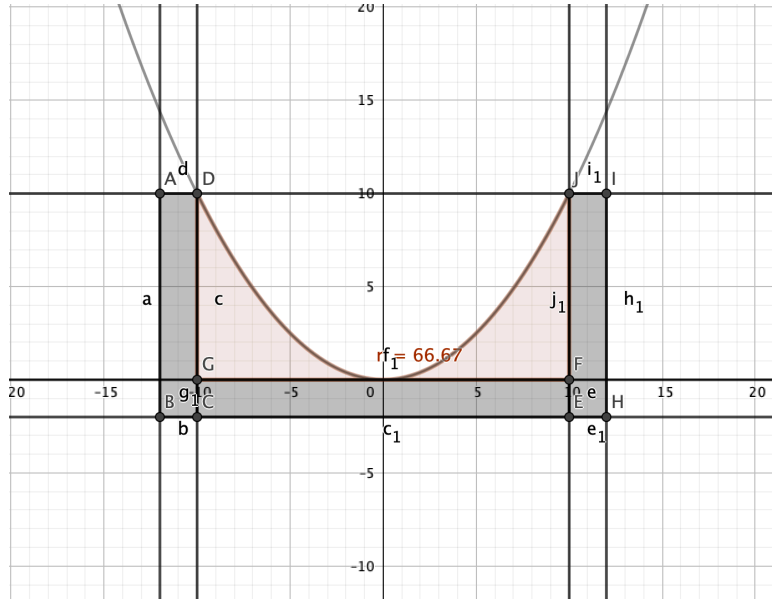


Figure 6: Cross Section represented by a quadratic function with degree 2 (Made by myself using GeoGebra)

$$Area = 2 \times \left(100 - \int_0^{10} \left(\frac{x^2}{10} \right) dx \right)$$

Using technology, we find the area:

$$Area \approx 133cm^2$$

Once again, to find the time taken to fill up the rain channel under the constant rainfall, we simply multiply the area by the 100cm length and divide by 2000 cubic centimetre per hour rainfall rate:

$$Time = \frac{133 \times 100}{2000} = 6.65hours$$

We can already tell that the quadratic cross section has a much lower flood time than the trapezoidal cross section, almost one hour less, even though they look quite similar.

If we want to find a function that calculates the time taken to fill up the rain channel for a given initial height h , we again manually integrate the function. To find what lower bound to use the integral, we once again need to find the x-intercept in terms of h :

$$y = \frac{x^2}{10} - h \rightarrow 0 = \frac{x^2}{10} - h \rightarrow x = \sqrt{10h}$$

We therefore plug in the upper and lower bounds:

$$\int_{\sqrt{10h}}^{10} \left(\frac{x^2}{10} - h \right) dx$$

Integrating, we get

$$\begin{aligned} & \left[\frac{x^3}{30} - hx \right]_{\sqrt{10h}}^{10} \\ Area &= 2 \times \left((100 - 10h) - \left[\frac{x^3}{30} - hx \right]_{\sqrt{10h}}^{10} \right) \\ Area &= 2 \times \left(100 - 10h - \left(\left(\frac{100}{3} - 10h \right) - \left(\frac{\sqrt{10h}^3}{30} - h\sqrt{10h} \right) \right) \right) \end{aligned}$$

Once again, this expansive formula simplifies to the formula:

$$Area = \frac{400}{3} + \frac{\sqrt{10h}^3}{15} - 2h\sqrt{10h}$$

Hence, to find the time taken to fill the rain channel of length 100cm, we multiply by 100 cm and divide by the rainfall rate of 2000 cubic centimetres per hour.

$$Time = \left(\frac{20}{3} + \frac{\sqrt{10h}^3}{300} - \frac{h\sqrt{10h}}{10} \right) hours \rightarrow 0 < h < 10$$

6. Semi-circular Rain Channel:

Rain channels can come in different shapes and sizes, which influences the volume of water they can hold. When looking at the rain channels in Tenerife, they looked like semi-circles, which is a typical shape for rain channels. In order to find the area of the semi circle using integration, we first must find a function which graph represents a semi-circle (see figure 7). On a graph, the following function represents a circle: $\sqrt{x^2 + y^2} = r$, where r is the radius of the circle. In order to integrate, we must rearrange for y. Rearranging for y, we arrive at the function:

$$\sqrt{x^2 + y^2} = r \rightarrow y = \pm \sqrt{r^2 - x^2}, -1 < x < 1, \\ x \in R$$

Note that we only take positive y-values, since we only need to find the area of half of the circle. Furthermore, by using this equation instead of one where the orientation like the actual rain channel, the manual integration later with varying initial water heights will be much simpler. Also, the area we are finding is still the same as if we were to flip the graph in order for it to resemble the rain channel. To find the area of the semi-circle, we can now find the definite integral of the function between the x values of 10 and -10, since this will give us the area under the semi circle:

$$\int_{-10}^{10} \sqrt{10^2 - x^2} dx$$

Using technology, we can this area to be:

$$Area = \int_{-10}^{10} \left(\sqrt{10^2 - x^2} \right) dx = 157cm^2$$

Assuming that the rain channel is empty when rainfall begins, we can multiply this area by the length of the rain channel, 100cm, to find the volume:

$$Volume = 157 \times 100 = 15,700cm^3$$

As from before, we can now divide this volume by the constant rainfall rate to find out how long it will take for the rain channel to fill up. We can still use the rate of 2,000 cubic centimetres per hour, since the area of the top of the rain channel is still 2000 centimetres squared, as before. Therefore, we calculate the time taken to fill the rain channel:

$$Time = \frac{15700}{2000} = 7.85hours$$

Now, lets consider cases where the rain channel already has a certain height of water in it before it starts to rain. To find the volume of 'empty' rain channel in terms of a given initial water

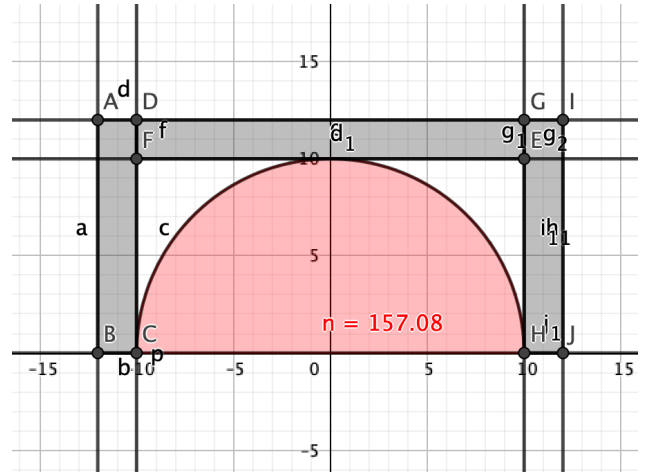


Figure 7: Semi-Circular Cross Section of radius 10, represented by the red area (Made by myself using GeoGebra)

height, we use integrals for the graph representing the cross section of the semi-circular rain channel. This scenario shows why simply using the formula for the area of a circle is limiting for this investigation, since we can only find the area of a semi circle, not of a part of a semi circle, as we are trying to find when looking at certain initial heights of water.

To make our integral easier to work with, let's find the area of one side of the cross-section, and then simply multiply this area by 2 to account for the entire cross-section. In the graph, the water already in the rain channel will start from the base, so at $x: 10$, and go to the $x: 10-h$, where h is the height of the water. Hence, the free area is the integral from $x: 0$ to $x: 10-h$, as seen in figure 8.

Considering this, we now have an expression for the area of the cross-section which is not filled up with water:

$$\int_0^{10-h} \left(\sqrt{10^2 - x^2} \right) dx$$

In order to manually integrate, we can use trigonometric substitution. By letting x be $10\sin\theta$, we get the new integral:

$$\int_0^{10-h} \left(\sqrt{10^2 - 10^2 \sin^2 \theta} \right) dx$$

From the trigonometric identities, the expression in the integral is equal to $10\cos\theta$.

Hence, we can replace $10\cos\theta$ in the integral:

$$\int_0^{10-h} (10 \cos \theta) dx$$

Since we now need to integrate with respect to θ , we differentiate $x = 10 \sin \theta$ on both sides, which turns the expression into $dx = 10 \cos \theta$. We can substitute this in the place of dx :

$$\int_0^{10-h} (100 \cos^2 \theta) d\theta$$

Since we are now differentiating for θ , we also need to change the upper and lower bounds by substituting them into $x=10\sin\theta$:

$$10 - h = 10 \sin \theta \rightarrow \theta = \arcsin \frac{10 - h}{10}$$

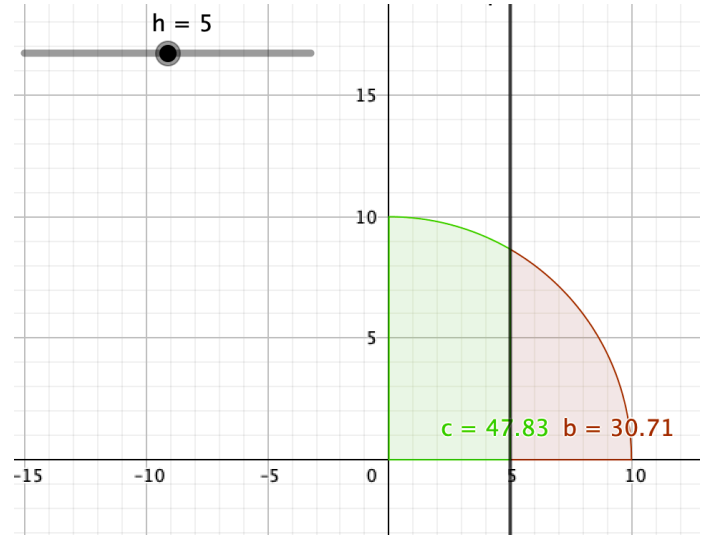


Figure 8: As initial height changes, so does the area filled by water, represented by the red region (Made by myself using GeoGebra)

$$0 = 10 \sin \theta \rightarrow \theta = \arcsin 0$$

$$100 \int_{\arcsin 0}^{\arcsin \frac{10-h}{10}} (\cos^2 \theta) d\theta$$

Using the trigonometric identity $\cos^2 \theta = \frac{\cos(2\theta) + 1}{2}$, we can substitute this into the integral:

$$50 \int_{\arcsin 0}^{\arcsin \frac{10-h}{10}} (\cos(2\theta) + 1) d\theta$$

Now, we can integrate the expression with respect to θ :

$$50 \left[\frac{\sin(2\theta)}{2} + \theta \right]_{\arcsin 0}^{\arcsin \frac{10-h}{10}}$$

Substituting the upper and lower bound will lead to the following expression:

$$50 \left(\frac{\sin \left(2 \arcsin \frac{10-h}{10} \right)}{2} + \arcsin \frac{10-h}{10} \right)$$

Since this gives us the area of half of the cross section, we multiply the expression by 2:

$$2 \times 50 \left(\frac{\sin \left(2 \arcsin \frac{10-h}{10} \right)}{2} + \arcsin \frac{10-h}{10} \right) = 100 \left(\frac{\sin \left(2 \arcsin \frac{10-h}{10} \right)}{2} + \arcsin \frac{10-h}{10} \right)$$

Once again, we set the length of the rain channel to be 1m, or 100cm. Also, since the area of the top of the rain channel remains unchanged, we divide the volume by 2000 cubic centimetres per hour, which is the rainfall rate. Multiplying the cross sectional area by 100cm and dividing by 2000, we arrive at the formula:

$$\frac{100}{2000} \times 100 \left(\frac{\sin \left(2 \arcsin \frac{10-h}{10} \right)}{2} + \arcsin \frac{10-h}{10} \right)$$

$$Time = \left(\frac{5}{2} \sin \left(2 \arcsin \frac{10-h}{10} \right) + 5 \arcsin \frac{10-h}{10} \right) hours \rightarrow 0 < h < 10$$

7. Comparing Time Functions for different Cross Sections:

We have now found a function for time taken to fill up a 20 cm wide, 10 cm deep and 100cm long rain channel in terms of the initial height of the water in the channel before it begins to rain. The graph shown in figure 9 compares each of the time functions for the four cross sections when the initial height of water is between 0 and 10 cm. Each of the lines represents the change in time taken to fill up the rain channel versus the change in initial height of water in the channel. It is

quite interesting to see how graphs of the trapezoidal and semi-circular cross section begin at similar points, then diverge and meet again at the point where h is 10. This shows that the trapezoidal cross section is more resembling of the semi-circular cross section than the quadratic, which is unexpected. The graph also shows that in fact, a semi-circular cross section is a much more optimal shape for a rain channel than the quadratic since it takes longer to fill up while still having the round shape.

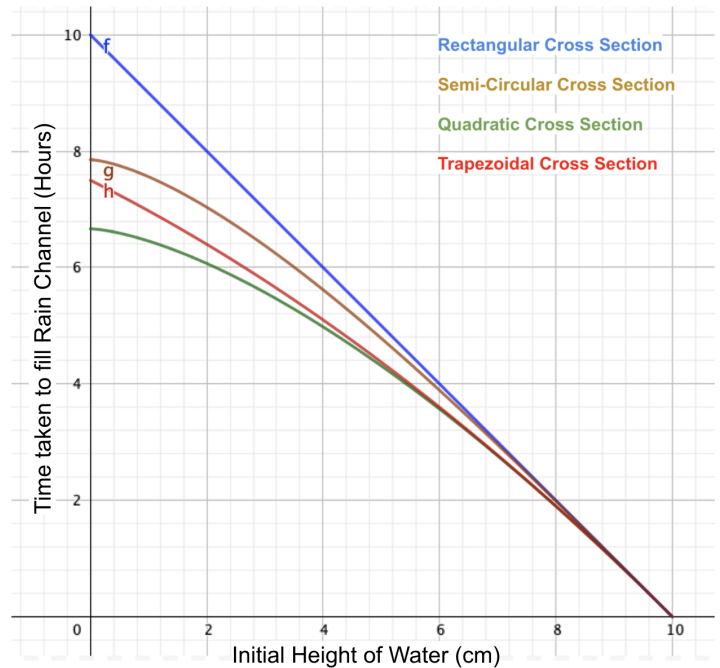


Figure 9: Graph showing relationship between initial height of water and time taken to fill different rain channels (Made by myself using GeoGebra)

8. Generalising the Time Function:

Up to this point, we have found 4 functions of time taken to fill up the rain channel in terms of the initial height of the water already in the rain channel. In these scenarios, we have set the length of the channel to be 100cm and we have set the rainfall rate to be constant. Furthermore, we have not considered the rain that would be entering the channel if the channel also received water from a street adjacent to it or from a roof it is attached to. Hence, we can find a function that has multiple parameters which can be changed to find the time taken for any given scenario. A visual representation of these different parameters can be seen in figure 10.

Let us first consider changing the length of the rain channel. For example, for the trapezoidal cross section, we set the length of the channel to be 100cm, and the rate of rainfall to be 2000 cubic centimetres per hour:

$$Time = \frac{100}{2000} \times \left(150 - 10h - \frac{h^2}{2} \right)$$

If we were to replace the 100cm length by a parameter, l , we could substitute this into the equation. However, the rate of water entering the rain channel depends on the length of the rain channel, since the rate of volume of rain is calculated by multiplying the 1 cubic centimetre per hour per square centimetre rate by the area of the top of the channel. Hence, as we change the length of the channel, so does the area of the top, and hence the rate of volume flowing in changes. We therefore substitute the area of the channel in for the rate of volume:

$$Time = \frac{l}{20 \times l} \times \left(150 - 10h - \frac{h^2}{2} \right)$$

In this case, we see that the length of the rain channel does not affect how long it takes to fill the channel, since as length increases, so does the amount of water entering the rain channel per hour.

Another scenario to consider is what changes in the time taken to fill the rain channel when we consider the width of a street next to the channel of which the rain falling onto it also flows into the rain channel. For the practicality of the calculations, let assume that the street is only slightly tilted towards the rain channel and so the

area of the street equals to the area which the rain will fall onto. We find the area of the street section that flows into the rain channel by multiplying the width, say w , by l , which is the length of the rain channel. This will be added to the area of the top of the rain channel, so we will add this into the equation as follows:

$$Time = \frac{l}{A_R + A_S} \times \left(150 - 10h - \frac{h^2}{2} \right)$$

$$Time = \frac{l}{(20 \times l) + (w \times l)} \times \left(150 - 10h - \frac{h^2}{2} \right)$$

Once again, if we factor out the parameter l from the denominator in the equation, the l from the numerator and the denominator cancel, so we are left with the equation:

$$Time = \frac{1}{20 + w} \times \left(150 - 10h - \frac{h^2}{2} \right)$$

We can now use this expression for the formulas describing the cross sectional area of all the other cross-sectional shapes looked at beforehand. The following equations calculate time for the different shapes:

Rectangular:

$$Time = \frac{1}{20 + w} \times (200 - 20h) \text{ hours}, 0 < h < 10$$

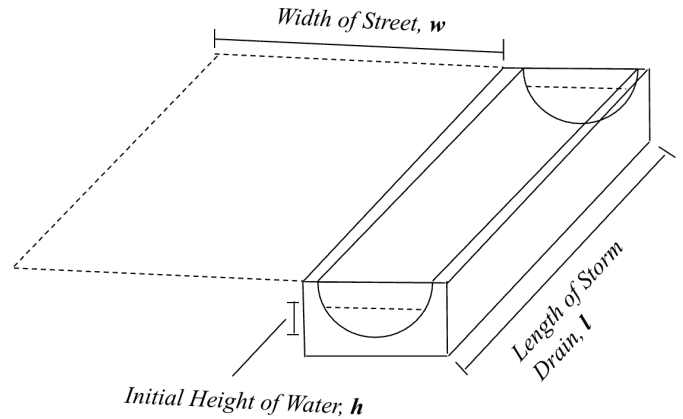


Figure 10: Sketch of the scenario where the rain channel drains street next to it, with different parameters labelled (Own Work)

Trapezoidal:

$$Time = \frac{1}{20 + w} \times \left(150 - 10h - \frac{h^2}{2} \right) \text{ hours}, 0 < h < 10$$

Quadratic:

$$Time = \frac{1}{20 + w} \times \left(\frac{400}{3} + \frac{\sqrt{10h}^3}{15} - 2h\sqrt{10h} \right) \text{ hours}, 0 < h < 10$$

Semi-Circular:

$$Time = \frac{1}{20 + w} \times 100 \left(\frac{\sin \left(2 \arcsin \frac{10-h}{10} \right)}{2} + \arcsin \frac{10-h}{10} \right) \text{ hours}, 0 < h < 10$$

Considering these equations, we can compare how long it takes to fill up each of the rain channels, taking into consideration that the rain channels drain the water from a 2m wide street as adjacent to them, hence, w is 200 cm, since 1 m is equal to 100 cm. Furthermore, let's assume the initial height of water in each of the channels is 2 cm, hence h is 2 cm. Using these parameters, we will find the time taken to fill up the channel for any given length of the rain channel, since we saw that the l terms in the formula cancel out. Using technology, we arrive at the following results:

$$\textbf{Rectangular: } Time = \frac{1}{20 + 200} \times (200 - 20 \times 2) = 0.727 \text{ hours} = 44 \text{ minutes}$$

$$\textbf{Trapezoidal: } Time = \frac{1}{20 + 200} \times \left(150 - 10h - \frac{2^2}{2} \right) = 0.582 \text{ hours} = 35 \text{ minutes}$$

$$\textbf{Quadratic: } Time = \frac{1}{20 + 200} \times \left(\frac{400}{3} + \frac{\sqrt{10 \times 2}^3}{15} - 2 \times 2\sqrt{10h} \right)$$

$$= 0.552 \text{ hours} = 33 \text{ minutes}$$

$$\textbf{Semi-Circular: } Time = \frac{1}{20 + 200} \times 100 \left(\frac{\sin \left(2 \arcsin \frac{10-2}{10} \right)}{2} + \arcsin \frac{10-2}{10} \right)$$

$$= 0.640 \text{ hours} = 38 \text{ minutes}$$

Naturally, once the part of the roof/ street which the rain channel is draining is taken into consideration, the times taken to fill the rain channels decrease significantly to less than an hour for all four of the different cross sections.

9. Flood times when rainfall rates are changing with time:

If we now consider the rainfall rate to be changing, say increasing at a constant rate every hour, we can use the suvat equations, which are originally equations describing motion in physics, to find the time taken to fill up the rain channels. The equation we will be focusing on is the following:

$$s = ut + \frac{1}{2}at^2$$

This describes the relation ship between distance, initial speed, acceleration and time. We will not go over how it is derived, but we know that it describes linearly changing velocities. However, we can apply this to our scenario by substituting volume for distance, the initial rainfall rate for initial speed, and the rate of increase of the rainfall for acceleration We can use this equation for our purposes since it is mathematically derived from the changing velocity graph over time. Velocity is just the derivative of distance with respect to time. The flow rate of rain we are looking at is similarly the derivative of volume with respect to time. Therefore we can use the equation to describe fill rate instead of speed and a constant change of fill rate instead of acceleration. We then arrive at the new equation:

$$V = rt + \frac{1}{2}at^2,$$

where V is the volume, r is the initial rainfall rate and a is the rate of increase of rainfall rate. The volume of the rain channel, which we found beforehand, is just the length of the channel times the cross-sectional area of the rain channel, so l times A . We can substitute this into the equation:

$$Al = rt + \frac{1}{2}at^2$$

Now let's consider the initial rainfall rate and rate of change of the rainfall rate. The initial rainfall rate we found earlier by multiplying the rate of 1 cubic centimetre per hour per square centimetre by the area of rain channel + area of the street the rain is falling on. We can substitute all of this in as

shown below:

$$r = 1 \times length \times width \rightarrow r = l \times (20 + w)$$

Substituting this in for r:

$$Al = (l \times (20 + w))t + \frac{1}{2}at^2$$

Lastly, the a term, which represents the constant rate of increase of the rainfall, will also depend on the area the rain is falling onto, multiplied by a constant which represents the increasing rainfall rate, say k . This is because the rate of increase of volume entering the channel once again is the increasing rate of rainfall, in this cubic centimetres per hour per square centimetre per hour, since the rate is changing, multiplied by the area the rain is falling onto, so l multiplied by the area. This will be easier to follow once we show it in the equation:

$$a = (A_{Street} + A_{Channel}) \times k$$

$$a = (l \times 20 + l \times w) \times k$$

$$a = l \times (20 + w) \times k$$

Substituting this in for a , we get:

$$Al = (l \times (20 + w))t + \frac{1}{2}(l \times (20 + w))kt^2$$

In order to simplify this equation, let us first factor out the $l \times (20 + w)$ part from both terms on the right side:

$$Al = (l \times (20 + w)) \left(t + \frac{1}{2}kt^2 \right)$$

If we now open the brackets around $l \times 20 + w$, we can divide both sides of the equation by l , which will get rid of the l term:

$$A = (20 + w) \left(t + \frac{1}{2}kt^2 \right)$$

We have now arrived at an equation that relates the cross-sectional area of the rain channel to the width of the street it is draining, the rate of increase of the rainfall, and the time taken to fill the rain channel. Note that the term A represents any of the four equations we have previously found for the area of the cross sections of differently shaped rain channels. Of course, we want to solve for time, but rearranging the equation for t would be complicated, since there is a t and a t squared term, and we can also just use technology to solve for t . Let us do a sample calculation to showcase the equation:

For the Area, let us use the formula for the rain channel with the semi-circular cross section. The formula for the area we found earlier, so let us substitute this in for A in the equation:

$$100 \left(\frac{\sin \left(2 \arcsin \frac{10-h}{10} \right)}{2} + \arcsin \frac{10-h}{10} \right) = (20 + w) \left(t + \frac{1}{2}kt^2 \right)$$

Once again, the h term represents the initial height of water in the rain channel. In order to let the increasing rainfall rate have an effect, let us set the initial height to a lower value of 2.5 cm. For the w term, a realistic value is 100 cm, about the width of a path or half of a small road. Lastly the k term represents the increase of the amount of cubic centimetres per square centimetre per hour, that the initial rainfall rate increases every hour. If the initial rainfall rate is 1 cubic centimetre per hour, a realistic increase is another 1 cubic centimetres every hour. This means that the rainfall steadily increases every hour by 1 cubic centimetre per hour. Substituting the three terms into the equation:

$$100 \left(\frac{\sin \left(2 \arcsin \frac{10 - 2.5}{10} \right)}{2} + \arcsin \frac{10 - 2.5}{10} \right) = (20 + 100) \left(t + \frac{1}{2} 1 t^2 \right)$$

Now, using technology, specifically the SolveN option on the calculator, we can solve for t , which stands for time taken to fill the channel:

t is 0.8 hours (48 minutes)

With a constant rainfall rate of 1 instead of an increasing rainfall rate, we arrive at a time of 1.12 hours, or 67 minutes, as shown below:

$$Time = \frac{1}{20 + 100} \times 100 \left(\frac{\sin \left(2 \arcsin \frac{10 - 2.5}{10} \right)}{2} + \arcsin \frac{10 - 2.5}{10} \right) = 1.12 hours$$

This means that if the rainfall increases with increments of 1 per hour, the time taken to fill the rain channel already decreases by close to a third. This shows that in a real world scenario where rainfall might become heavier or lighter over time we can use the equation to find the approximate time it will take for the rain channel to flood.

Conclusion:

Overall, this investigation has shown how the shapes of rain channels affect how quickly they fill up as well as how this time taken for them to fill up changes as the initial height of water already in the channel changes. Naturally, the rectangular rain channel takes the longest to fill up since it utilises the maximum volume in the rain channel with the dimensions we set for the investigation. We have also found that the semi-circular rain channel is the optimal shape since it has the property of funnelling water towards the middle, as do the trapezoidal and quadratic cross-sections, but it takes longer to fill up than the other two mentioned cross sections and furthermore the graph comparing time to initial height of water proved that for small initial heights of water, the time taken to fill the rain channels decreased the least for the semicircular rain channel, showing it is the optimal cross-section for a rain channel, as seen in figure 11.

Furthermore, through manual integration we have been able to construct equations for each of the shapes of rain channels which have different adjustable parameters which can be used to find the fill times for a variety of scenarios where initial height of water or the width of the street which the channel is draining changes. This has real-world significance as when planning the constructions of paths or streets these kind of equations are necessary to see how large rain channels have to be to avoid flooding in certain rain conditions or how high the risk of rain channels flooding is.

Furthermore, through the manipulation of the SUVAT equations of motion used in physics, we were able to incorporate changing rainfall rates into the

parameters to see how much an increasing rainfall rate reduces the time taken to fill up the channel. We found that just a slowly increasing rainfall rate can significantly reduce the time taken to fill up the rain channel. Incorporating the possibility of changing rates of rainfall into the investigation further makes it applicable to real life, where rainfall rates will constantly change over time.

It would have been interesting to see how water levels in the channels increase or decrease when water leaves the rain channels at different rates to see whether the channels will flood or not. Furthermore, as an extension to this investigation, one could look at cases where the rain channels have certain inclines and see how this relates to the speeds at which water leaves the channels, looking at forces of gravity and working with aspects of trigonometry.

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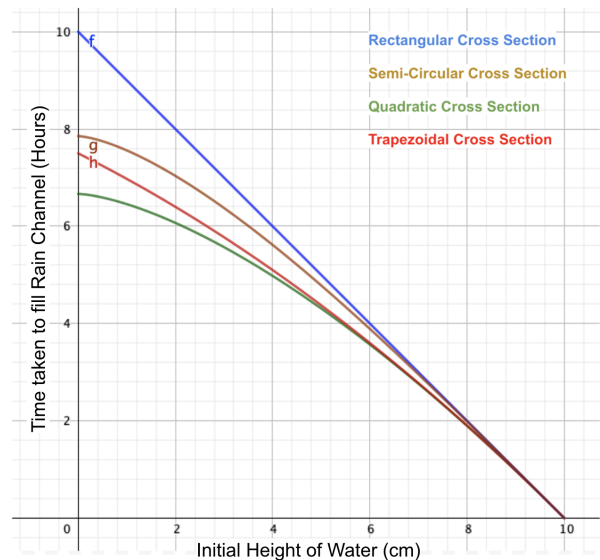


Figure 11: Graph showing relationship between initial height of water and time taken to fill different rain channels (Made by myself using GeoGebra)